

Detecting drought stress from the leaf to the landscape: is canopy structure or leaf desiccation more influential in tree canopy spectral responses to declining water?

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Summary

- Remotely sensed measurements of equivalent water thickness (EWT) derived from visible–shortwave infrared imaging spectroscopy data have the potential to revolutionize how we monitor drought in natural ecosystems. However, the mechanistic underpinnings of EWT measurements are still relatively unknown.
- We collected coincident measurements of leaf spectra, leaf water content, and leaf water potential over the course of a tabletop leaf drydown for 13 tree species. We then used a vegetation radiative transfer model to scale these measurements from the leaf to canopy scale, simulating how a leaf drying event would appear from an airborne or satellite instrument.
- We found that EWT responses to changes in leaf water content are strongly species-specific at both the leaf and canopy level. Furthermore, changes in EWT associated with declining leaf water are relatively small compared to those associated with potential canopy structural changes, such as leaf shedding or shifting leaf angle distribution, and with retrieval errors of current instruments.
- Our results suggest that single point-based EWT measurements are unlikely to reliably detect drought stress across species due to several confounding factors, but that time series data (newly available through operational and upcoming spaceborne imaging spectrometers) may enable approaches that circumvent many of these challenges.

Introduction

The emerging tools ecophysicists leverage to detect and quantify water stress in the field have added significantly to our understanding of plant hydraulic processes and how they break down during drought (Drew & Downes, 2009; Poyatos *et al.*, 2021; Humphrey & Frankenberg, 2023). However, moving beyond organismal understanding and scaling drought risk over a larger landscape is challenging, and our ability to map and monitor tree water stress at the stand or regional scale is comparatively much more limited. This data deficiency has downstream consequences: Despite significant advances in our understanding of links between climate and plant water stress, our ability to identify vulnerable trees and forecast mortality remains relatively limited (De Kauwe *et al.*, 2020; Trugman *et al.*, 2021; Venturas *et al.*, 2021).

Visible–shortwave infrared (VSWIR) imaging spectroscopy could potentially expand our ability to study and respond to plant drought at landscape to regional scales, but also introduces its own complications. VSWIR imaging spectrometers are passive optical instruments that measure reflected radiance in hundreds of narrow wavelength bands. This high density of information

provides a very sensitive means by which to probe leaf and canopy properties (Asner, 1998), including symptoms of tree water stress. Leaf and canopy reflectance both change when plants experience water stress (Hunt & Rock, 1989; Cibula *et al.*, 1992; Stefanski *et al.*, 2025), and a number of studies have successfully leveraged this signal to study canopy hydration over large areas (Dennison *et al.*, 2003; Miller *et al.*, 2022). However, spectral responses to stress can be extremely diverse, due to both the diversity of stress responses among tree species (Rennenberg *et al.*, 2006) and differences in how those stress responses affect leaf and canopy optical properties (Brodrick & Asner, 2017; Boving *et al.*, 2025). This aligns with results from other problems in imaging spectroscopy: Empirical models relating surface reflectance to vegetation properties, such as vegetation fractional cover or leaf foliar traits, generally perform best when calibrated with local measurements (Elmore *et al.*, 2000; Liu *et al.*, 2021; Wang *et al.*, 2023; Shen *et al.*, 2025). These factors make it challenging to apply imaging spectroscopy analyses outside of ecologically, spatially, and temporally narrow training domains or to connect spectral changes with any specific tree drought response.

One avenue that attempts to strike a balance between physical and empirical approaches is measurements of canopy equivalent

water thickness ($\text{EWT}_{\text{canopy}}$) (Allen *et al.*, 1971; Tucker, 1980). $\text{EWT}_{\text{canopy}}$ (sometimes referred to as canopy water content, or CWC) is, in theory, a physical property: It measures the distance that light traveled through liquid water contained in vegetation on its path from the sun to a sensor. Thus, $\text{EWT}_{\text{canopy}}$ should have a tight relationship with canopy-level leaf water per unit ground area ($\text{LWA}_{\text{canopy}}$). $\text{LWA}_{\text{canopy}}$ can in turn be connected with a specific tree water stress response: changing leaf water content with changes in leaf water potential – the motive force moving water through the liquid phase of the soil–plant–atmosphere continuum. Leaf water potential is in part indicative of environmental conditions, particularly soil moisture and evaporative demand, and regulated by the plant via the stomata that close to mitigate or prevent dieback and mortality (Hall & Kaufmann, 1975; Klein *et al.*, 2014; Bartlett *et al.*, 2016; Martin-StPaul *et al.*, 2017). However, there are a number of key uncertainties in how $\text{EWT}_{\text{canopy}}$ relates to $\text{LWA}_{\text{canopy}}$ in practice, making it hard to operationalize. First, leaf spectral responses to drying are diverse, with several existing papers on the topic yielding different or even conflicting results (Labovitz *et al.*, 1984; Carter, 1991; Aldakheel & Danson, 1997; Yu *et al.*, 2000). It is unclear whether or not these diverging spectral responses also affect leaf-level measurements of EWT (EWT_{leaf} , per unit leaf area). Further, VSWIR light traces nonlinear paths through a canopy, transmitting through and reflecting off of leaves. As a result, any surface reflectance measurement is affected not only by leaf spectral reflectance but also by canopy structure (Myneni *et al.*, 1989; Asner, 1998; Bowyer & Danson, 2004; Ollinger, 2011), which is itself responsive to water stress, as stressed trees can modulate their leaf angle or shed their leaves (Ehleringer & Comstock, 1987; Munné-Bosch & Alegre, 2004). The complexity of canopy optical processes means that these stress responses could not only directly affect $\text{EWT}_{\text{canopy}}$ but also change the relationship between $\text{EWT}_{\text{canopy}}$ and $\text{LWA}_{\text{canopy}}$, introducing additional challenges in interpreting patterns in $\text{EWT}_{\text{canopy}}$ across space and time.

Many of the studies that have attempted to measure the relationship between $\text{EWT}_{\text{canopy}}$ and $\text{LWA}_{\text{canopy}}$ have used a ‘top-down’ approach, comparing satellite or airborne $\text{EWT}_{\text{canopy}}$ measurements to field measurements of plant water status, with mixed results. For example, there have been several studies that collected *in situ* data on leaf water content and/or leaf biomass and attempted to directly relate it to $\text{EWT}_{\text{canopy}}$ (Roberts *et al.*, 1997; Ceccato *et al.*, 2001; Cheng *et al.*, 2008; Colombo *et al.*, 2008; Martin *et al.*, 2018; Raj *et al.*, 2021; Wong *et al.*, 2023; Allen *et al.*, 2026). These studies include many different species and ecosystems, ranging from corn fields to giant sequoia forests, and have produced similarly diverse results, encompassing both strong and null correlations. There have also been studies that approached the problem indirectly, comparing measurements of $\text{EWT}_{\text{canopy}}$ to tree mortality rates, climatic stressors, or canopy temperature (Asner *et al.*, 2015; Brodrick & Asner, 2017; Brodrick *et al.*, 2019; Baeza *et al.*, 2021; Miller *et al.*, 2022). These less direct approaches that examine changes in $\text{EWT}_{\text{canopy}}$ and relate them to other signals that we would expect to observe in a forest experiencing water stress have yielded

promising results, but do not necessarily illuminate the mechanisms that drive the underlying relationship between mortality and $\text{EWT}_{\text{canopy}}$ at the organismal level. There are several potential candidate mechanisms that could drive a relationship between $\text{EWT}_{\text{canopy}}$ and plant water stress that could take place concurrently with changing $\text{LWA}_{\text{canopy}}$ but are not mechanistically related to LWA_{leaf} , such as early leaf senescence (Zhang *et al.*, 2025) or dieback of understory plants (Wright, 1991; Tng *et al.*, 2022). Overall, while the theoretical basis for using $\text{EWT}_{\text{canopy}}$ to monitor tree water stress is strong, the evidence is much more mixed.

In this paper, we hypothesize that the discord in the published research on $\text{EWT}_{\text{canopy}}$ is a direct consequence of the complex optical interactions between light and vegetation that are pervasive in the VSWIR. We demonstrate that tree drought responses that take place concurrent with changing $\text{LWA}_{\text{canopy}}$, such as changes in leaf angle and leaf shedding, directionally confound the relationship between $\text{LWA}_{\text{canopy}}$ and $\text{EWT}_{\text{canopy}}$. To test our hypothesis, we collected spectral measurements of drying leaves to assess how declining leaf water manifests spectrally at the leaf level. Then, we used a radiative transfer modeling framework to assess how spectral changes in leaves associated with dehydration propagate to canopy scales and manifest in airborne or satellite remotely sensed spectra. Specifically, we ask (1) How do VSWIR leaf spectra change as a leaf dries? What leaf traits mediate spectral changes? (2) How do spectral responses to declining water propagate up in biological scales of organization from leaf to canopy-level measurements of $\text{EWT}_{\text{canopy}}$? Are the effects of changing leaf spectra with desiccation large or small relative to canopy structural changes, such as changes in leaf angle and leaf area, that can occur during the plant response to water limitation? (3) What are the implications of these results for operationalizing tree water stress detection with imaging spectroscopy using currently available instrumentation?

Materials and Methods

Study species and measurement procedure

Study species were selected to maximize variation in the relationship between spectral reflectance and leaf water per area (LWA) that could be observed in a realistic remote sensing scenario, including differences in leaf structure and drought response strategies. Previous research has identified interactive effects between leaf mass per area (LMA) and absorption of light by liquid water (Ceccato *et al.*, 2001); to measure this effect and compare it with other factors, we selected species that covered a wide range of LMA values (from *Populus trichocarpa*, 7.3 mg cm^{-2} to *Heteromeles arbutifolia*, 23.9 mg cm^{-2} ; see Table 1). We also targeted species with diverse hydric strategies, including different leaf habits (evergreen vs deciduous) and stomatal responses to drought (relatively ‘isohydric’ vs ‘anisohydric’), to maximize the overall leaf structural variability in our dataset. Next, to maximize the utility of the results for a broad audience of plant physiologists and remote sensing scientists, we targeted species with growth forms that could be reasonably identified in high- to

Table 1 Description of study species.

Scientific name	ID	Common name	No. of leaf samples	LMA (mg cm ⁻²)	TLP (MPa)
<i>Acer saccharinum</i> L.	ACSA	Silver Maple	6	8.1 ± 0.4	-2.18 ± 0.30
<i>Citrus limon</i> L.	CILI	Lemon Tree	3	15.8 ± 0.7	-2.43 ± 0.15
<i>Eucalyptus camaldulensis</i> Dehnh.	EUCA	River Red Gum	3	16.7 ± 0.8	-2.61 ± 0.06
<i>Eucalyptus globulus</i> Labill.	EUGL	Blue Gum	3	22.5 ± 1.8	-1.87 ± 0.04
<i>Eucalyptus sideroxylon</i> A.Cunn. ex Woolls	EUSI	Red Ironbark	6	12.3 ± 0.8	-2.16 ± 0.11
<i>Heteromeles arbutifolia</i> (Lindl.) M.Roem.	HEAR	Toyon	6	23.9 ± 4.6	-4.16 ± 0.31
<i>Malus domestica</i> (Suckow) Borkh.	MADO	Apple	2	8.6 ± 0.4	-2.43 ± 0.00
<i>Pachygone laurifolia</i> (DC.) L.Lian & Wei Wang	PALA	Laurel-leaved Snail Tree	3	17.5 ± 4.2	-2.89 ± 0.37
<i>Persea americana</i> Mill.	PEAM	Avocado	4	13.1 ± 1.2	-2.37 ± 0.37
<i>Platanus racemosa</i> Nutt. ex Audubon	PLRA	Western Sycamore	4	9.7 ± 1.1	-2.22 ± 0.23
<i>Populus trichocarpa</i> Torr. & A.Gray ex Hook.	POTR	Black Cottonwood	3	7.3 ± 1.5	-3.09 ± 0.76
<i>Quercus agrifolia</i> Née	QUAG	Coast Live Oak	4	15.6 ± 0.8	-2.76 ± 0.24
<i>Quercus douglasii</i> Hook. & Arn.	QUDO	Blue Oak	2	14.1 ± 1.3	-3.37 ± 0.13

Dry leaf mass per area (LMA) and turgor loss point (TLP) are expressed as the mean value measured for that species in this study, plus/minus one SD.

mid-resolution hyperspectral remotely sensed data. Due to limitations in leaf spectroscopy instrumentation, we were only able to measure broadleaf species. Finally, we sampled across multiple plant families to capture taxonomic diversity, including both native and non-native species within our study area of Santa Barbara, California. Altogether, we collected data for 13 species that covered a range of LMA values and drought tolerances (Supporting Information Fig. S1; Table 1), including large trees and shrubs from 9 different genera and 8 different families.

To measure how leaf spectra change as leaves dry, we used a pressure-volume curve (P-V curve) procedure (Tyree & Hammel, 1972) modified to include measurements of leaf spectral reflectance. P-V curves are collected by gradually drying a leaf in ambient air while taking near-concurrent measurements of leaf water mass and leaf water potential. As the leaf dries over several hours, water naturally evaporates out of the leaf, simulating a natural leaf drying event. Concurrent measurements of leaf water mass and leaf water potential throughout this procedure provide insight into how water potential changes as a leaf dries and can be used to calculate several leaf traits that are key to understanding water relations in the leaf. In the early stages of a drydown, leaf water potential and mass changes are driven primarily by changes in turgor pressure potential, with water potential often declining rapidly with small changes in water mass. Eventually, the leaf dries enough to lose turgor, after which water potential is driven by the osmotic potential of concentrating solutes and tends to decline less rapidly. The leaf water potential associated with this regime change is called the turgor loss point (TLP), sometimes also called the wilting point, and represents the degree of leaf dryness beyond which leaf cells lose turgor and leaf function rapidly declines (Sack *et al.*, 2003). In addition to influencing the relationship between leaf water mass and leaf water potential, TLP as a trait varies across species and is associated with hydraulic vulnerability (Blackman *et al.*, 2010) and stomatal strategy (Brodrick & Holbrook, 2003; Lenz *et al.*, 2006).

For each of the 13 study species, we collected sun-exposed leaves from one representative tree and gradually dried them out on a benchtop, taking near-concurrent measurements of leaf

water potential (using a Scholander-type pressure chamber; PMS Instruments, Albany, OR, USA), leaf water content (using a Sartorius Entris II balance), and leaf spectral reflectance (using an ASD FieldSpec 4). Based on a more intensive sampling of multiple individuals from three different species, intraspecific variability across individual trees in leaf spectral responses to drying was low relative to interspecific variability (Fig. S2); therefore, for further analysis, we targeted sampling multiple leaves per individual rather than many individuals per species to maximize the diversity across species, as this analysis is intended to assess how leaf spectral changes are mediated by leaf traits. We collected enough replicates for each tree/species to constrain the SE of the species' mean TLP to within 0.5 MPa, typically 3–4 leaves. Leaves were collected shortly before sunrise to maximize the starting hydration level and eliminate the need for leaf rehydration, and all leaves had an initial water potential at least 1 MPa less negative than their turgor loss point (Fig. S3). We stored leaves in a cool, high-humidity environment for no > 72 h before measurement. Leaf water potential, followed immediately by leaf mass and then leaf reflectance, was repeatedly measured on drying leaves (see Fig. S4 for annotated lab setup diagram). The location and orientation of the leaf clip were marked on each leaf to ensure consistent view area and angle throughout the curve. After the P-V curve procedure, leaf area was measured using a digital scanner and IMAGEJ (Schneider *et al.*, 2012), and dry mass was measured after 72 h in a drying oven. Spectral reflectance measurements were resampled to a spectral resolution of 1 nm using cubic spline interpolation. Finally, we used the pvcurve PYTHON package v.0.1.0 (Allen, 2026) to calculate LWA, TLP, and several other leaf water relations traits that may mediate the relationship between leaf spectral reflectance and LWA_{leaf}, including bulk elastic modulus, osmotic potential at full turgor, apoplastic water fraction, and saturated water content.

Spectra processing

To compare leaf spectral responses to leaf drying across our study species, we calculated three spectra-derived values for each leaf

spectrum: two classic vegetation indices, the normalized difference vegetation index (NDVI) and the normalized difference moisture index (NDMI), and the physically derived EWT. NDVI, NDMI, and EWT have a long history of use for assessing vegetation health and hydration. NDVI was developed for use in mapping vegetation abundance with early satellite and airborne instruments. NDVI is calculated as a normalized difference of the near-infrared and red reflectance measurements (Kriegler *et al.*, 1969):

$$\text{NDVI} = \frac{\rho_{895} - \rho_{675}}{\rho_{895} + \rho_{675}} \quad \text{Eqn 1}$$

where ρ_λ indicates the reflectance measured at wavelength λ . NDMI was similarly formulated as a proxy for vegetation moisture content and is calculated using near-infrared and shortwave-infrared bands (Hardisky *et al.*, 1983):

$$\text{NDMI} = \frac{\rho_{820} - \rho_{1600}}{\rho_{820} + \rho_{1600}} \quad \text{Eqn 2}$$

Leaf-level NDVI and NDMI ($\text{NDVI}_{\text{leaf}}$ and $\text{NDMI}_{\text{leaf}}$, respectively) were calculated for each individual spectral measurement so that changes in $\text{NDVI}_{\text{leaf}}$ over the drydown could be compared across samples. While NDVI and NDMI measure no specific vegetation property, empirical relationships can be built between these indices and many useful plant traits and fluxes, making them common and valuable tools in many remote sensing studies.

By contrast, EWT is formulated to directly measure the amount of leaf water per unit area and can be calculated in a variety of different ways depending on the use case and data availability. We calculate leaf-level EWT (EWT_{leaf}) using a Beer–Lambert approach, as implemented as an accessory module in ISOFIT v2.9.9.2 (Brodrick *et al.*, 2024). This definition of EWT is commonly used in remote sensing studies, as it is typically retrieved simultaneously with many popular surface reflectance retrieval algorithms (Green *et al.*, 2006; Bohn *et al.*, 2020) and therefore comes at little to no additional computational cost. EWT_{leaf} is defined as the value for EWT that maximizes the fit between a measured and modeled reflectance ρ_λ , where the modeled reflectance ρ_λ is defined as:

$$\rho_\lambda = (a + b\lambda) e^{-\alpha_\lambda \cdot \text{EWT}} \quad \text{Eqn 3}$$

where α_λ is the absorption coefficient of liquid water at wavelength λ , and a and b are empirical parameters that account for loss of light to other sources, such as leaf transmittance. To maximize the number of measurements used to constrain EWT_{leaf} while avoiding wavelengths affected by the spectrometer discontinuity present in the ASD FieldSpec 4 instrument, we used leaf reflectance measurements from the wavelength range of 1120–1280 nm to calculate EWT_{leaf} . These tools – NDVI/NDMI and EWT – complement each other and have all seen diverse applications in studies of leaf spectra, especially for assessing vegetation health and measuring leaf water.

Leaf-level statistical analyses

To determine how leaf drying affected individual reflectance bands and/or spectral properties (such as $\text{NDVI}_{\text{leaf}}$, $\text{NDMI}_{\text{leaf}}$ or EWT_{leaf}), we fit a series of linear mixed-effects models. Each model (M) was fit multiple times using all spectral measurements as response variables – 2151 different spectral bands plus 3 models of derived variables (as $\text{NDVI}_{\text{leaf}}$, $\text{NDMI}_{\text{leaf}}$ and EWT_{leaf}) for a total of 2154 fits per model version. Said another way, the response variable Y was either $\text{NDVI}_{\text{leaf}}$, $\text{NDMI}_{\text{leaf}}$, EWT_{leaf} or one of the 2151 ρ_λ reflectance bands. All models were fit in R v.4.3.2 (R Core Team, 2021) using lme4 (Bates *et al.*, 2015) and lmerTest (Kuznetsova *et al.*, 2017).

First, we modeled each response variable Y as a function of LWA_{leaf} including each leaf sample's unique identification number (ID) as a random effect to separate within-sample and cross-sample variation:

$$\text{M0} : Y \sim 1 + \text{LWA}_{\text{leaf}} + (1|\text{ID})$$

To assess how turgor loss affected leaf reflectance, for models with reflectance as the response variable, we further separated M0 into pre-TLP ρ_λ and post-TLP ρ_λ . Next, to determine how changes in LWA_{leaf} relate to changes in each spectral response variable, we explicitly modeled the change in each response variable as a function of the change in LWA_{leaf} :

$$\text{M1} : \Delta Y \sim 1 + \Delta \text{LWA}_{\text{leaf}} + (1|\text{ID})$$

where ΔY and $\Delta \text{LWA}_{\text{leaf}}$ are the change in Y and LWA_{leaf} respectively, for each measurement and sample from that sample's estimated value at leaf water saturation, calculated using linear regressions between each variable and relative water content for all measurements where leaf cells had not yet lost turgor. Finally, to determine what factors moderate the relationship between LWA_{leaf} and leaf spectral reflectance, we modeled each response variable as a function of LWA_{leaf} as well as several water relations traits, including species as an additional random effect:

$$\text{M2} : Y \sim 1 + \text{LWA}_{\text{leaf}} + \text{LMA} + \text{SWC} + \text{AWF} + \Psi_{\text{sFT}} + \epsilon + (1|\text{species}/\text{ID}).$$

where SWC is saturated water content, AWF is apoplastic water fraction, Ψ_{sFT} is osmotic potential at full turgor, and ϵ is bulk tissue elastic modulus. Each of these water relations traits was scaled so as to provide comparable estimates of effect size. To contextualize the relative importance of each of these water relations traits, we also calculated correlation coefficients for linear regressions between each possible combination of traits.

To determine if TLP represented a regime change in spectral responses to drying, we conducted Chow tests (Chow, 1960) using the chowtest python package (Clements, 2025) for all samples and all wavelengths, regressing reflectance at each wavelength against LWA_{leaf} and considering the LWA_{leaf} at TLP as a breakpoint.

Scaling from leaf measurements to canopies

To scale our spectral data from leaf-level measurements to remotely sensed measurements, we simulated vegetation canopy reflectance using the vegetation radiative transfer model PROSPECT+SAIL, or PROSAIL (Baret *et al.*, 1992). PROSAIL has two components: PROSPECT (leaf optical PROPERTIES SPECTra) (Jacquemoud & Baret, 1990), which simulates leaf spectral measurements, and SAIL (Scattering by Arbitrary Inclined Leaves) (Verhoef, 1984, 1985), which uses outputs from PROSPECT to simulate canopy reflectance that would be measured by an airborne or spaceborne instrument. We selected PROSAIL because it is well-studied and well-validated for the specific vegetation traits that will likely have the strongest effects on how spectral measurements are affected by plant water stress – specifically, LMA, LWA, leaf area index (LAI), and leaf angle distribution. These simulations enabled us to scale leaf-level processes such as leaf orientation and leaf traits and understand how they integrate to affect EWT_{canopy} and LWA_{canopy} (the actual foliar canopy water content per unit area, calculated as LWA_{leaf} times LAI).

We propagated our leaf-level spectral measurements through PROSAIL to scale organ-level processes and understand how each physiological factor individually during a drydown event might manifest at the canopy scale in remotely sensed data for a variety of canopy structures. First, we ran the PROSPECT model in inverse mode to retrieve the parameters that best corresponded to our spectral measurements. This inversion was performed using only the wavelengths of light used in an EWT_{leaf} / EWT_{canopy} calculation (1120–1280 nm) to ensure maximum accuracy at these wavelengths. Note that the parameters retrieved for PROSPECT are completely incidental to the analysis presented here: We use PROSPECT as a means to represent our measured spectra in the radiative transfer modeling framework, and not to retrieve any leaf properties. Estimates of EWT_{leaf} derived from the simulated spectra were biased slightly high when compared to measured spectra (mean bias = 0.002 g cm^{-2}); however, this bias was small, relatively consistent across species, and invariant over the course of the drydown. Regardless, to mitigate the effects of this bias on our results and inferences, we focus principally on changes in EWT_{canopy} rather than the absolute value.

Next, we simulated three different kinds of idealized drydowns intended to capture the range of values observed for: (1) leaf-level drydowns, (2) canopy structural drydowns (e.g. changes in leaf orientation), and (3) combined leaf and structural drydowns. For the leaf drydown, we used SAIL to simulate changes in EWT_{canopy} associated with declines in leaf water using our leaf spectral measurements. For the structural drydown, we simulated increases in mean leaf angle (from 30° at the start of the drydown to 45° at the end, modeled as a linear function of LWA_{leaf}) typically associated with wilting or sun aversion that has been observed to occur during drought in multiple species (Ehleringer & Comstock, 1987) while maintaining leaves at full saturated water content for simulation purposes. Finally, for the leaf and structural drydown, we simultaneously simulated declining leaf

water and increasing mean leaf angle. Using SAIL, we created a lookup table of canopy spectral measurements that covered the relevant parameter space for these simulations, and interpolated between the parameters with linear regressions. This set of simulations is not designed to capture any specific plant's responses to drying, but rather to allow us to separate out and attribute the interacting effects of leaf spectral changes and canopy structural changes that have been previously documented at the leaf and canopy level in trees under drought conditions.

Care should be taken when translating our results to real-life ecosystems, which exhibit heterogeneity that is unaccounted for in our idealized modeling framework. For example, in this paper, we simulated homogeneous canopies where each pixel has a green vegetation fraction of 100%. However, we would expect different EWT_{canopy} measurements between a pixel with homogeneously low LAI and a pixel with heterogeneous patches of moderate LAI and bare soil, even if the two pixels had the same average LAI values. While the assumed homogeneity in our idealized model experiments is a reasonable approximation in large contiguous forests, in many ecosystems with open or patchy canopy structures (such as savannas, mixed woodlands, or orchards), this assumption holds less well.

Calculating EWT measurement sensitivity

Finally, to benchmark the size of the leaf drydown signal against realistic sensor noise, we estimated the error introduced to EWT_{canopy} measurements during realistic conditions by simulating data representative of current operational instruments and applying standard data processing procedures. Measurements from airborne or satellite sensors can be essentially decomposed into three components – (1) the sensor characteristics, (2) atmospheric effects, and (3) the surface reflectance. Typically, the sensor characteristics are well-known. By contrast, atmospheric effects (i.e. light attenuation and scattering) vary across space and time due to differences in aerosol concentration, water vapor, or other properties. These effects must be removed to accurately characterize the surface reflectance, which can be related to vegetation traits and properties. While errors introduced through this process are small using modern methods, these errors could be non-negligible when characterizing subtle spectral changes, such as those we would expect to observe during leaf dehydration.

To calculate EWT_{canopy} measurement error, we simulated top-of-atmosphere imaging spectrometer data under diverse viewing and atmospheric conditions for two currently operational instruments: Airborne Visible InfraRed Imaging Spectrometer – Next Generation (AVIRIS-NG) and the Earth Surface Mineral Dust Source Investigation (EMIT) instrument. AVIRIS-NG (Hamlin *et al.*, 2011) is an airborne imaging spectrometer frequently deployed for terrestrial science applications, and EMIT (Green *et al.*, 2020) is a spaceborne imaging spectrometer housed on the international space station. Both instruments offer advantages that could contribute to reducing EWT_{canopy} measurement error: AVIRIS-NG has a higher spectral resolution than EMIT (AVIRIS-NG: 5 nm, EMIT: 7.4 nm) but a generally lower measurement precision due to several technological advancements

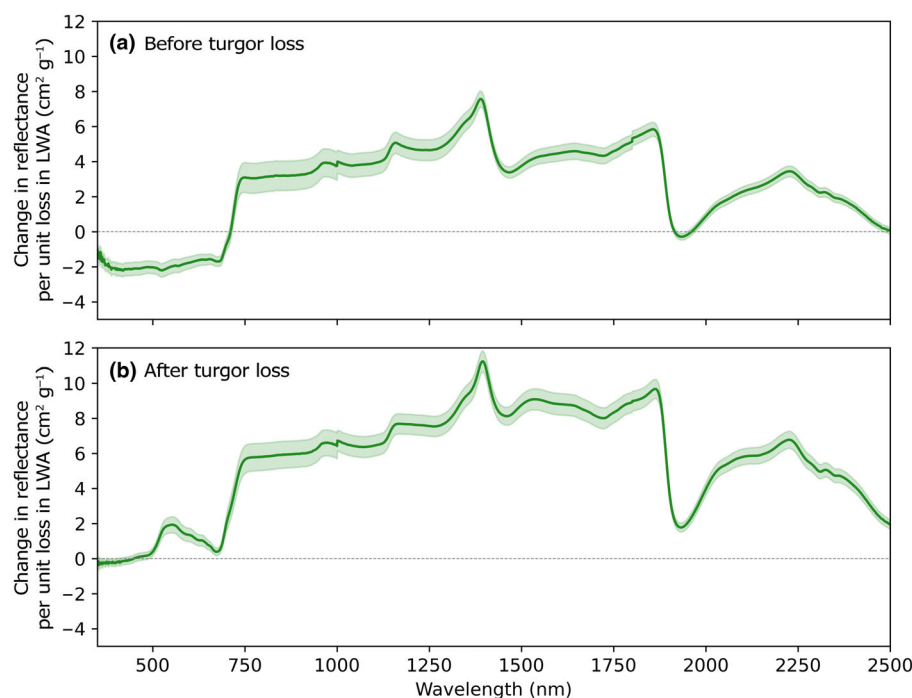


Fig. 1 On average, drying leaves became less reflective at visible wavelengths up until turgor loss, and more reflective at near-infrared wavelengths over the entire drydown. Values on the y-axis indicate the slope between leaf water per area (LWA) and leaf reflectance as estimated in M0; in this graph, positive values on the y-axis indicate an increase in leaf reflectance as the sample dries, and negative values indicate a decrease in reflectance. The green line shows the slope of the LWA effect in M0 at each wavelength (a) before turgor loss and (b) after turgor loss; the shaded area shows a 95% confidence interval. For a list of species considered in this analysis, see Table 1.

that took place in the years between the two instruments' development. We simulated measurements from each sensor for a diverse range of vegetation species and types by leveraging surface reflectance measurements in the ECOSTRESS spectral library (Meerdink *et al.*, 2019) and a diverse range of realistic viewing and atmospheric conditions using the Surrogate Radiative Transfer Modeling Network (sRTMnet) (Brodrick *et al.*, 2021). By comparing EWT_{canopy} values calculated from the raw surface reflectance data with EWT_{canopy} values calculated using our simulated dataset, we can establish the error introduced through instrumentation and the surface reflectance retrieval process. More detail on our process is available in Notes S1.

Interestingly, we found no relationship between the magnitude of the EWT_{canopy} error and any of the considered viewing conditions or atmospheric parameters – instead, EWT_{canopy} error values varied primarily with respect to the instrument being used, and partly on the magnitude of EWT_{canopy} , with lower EWT_{canopy} values being associated with higher error (Fig. S5). Therefore, we used the outputs of our analysis to establish a single average EWT_{canopy} error value for each instrument, including both the bias and precision of the measurement for each of the instruments considered.

Results

Leaf spectral responses to drydowns varied across species

All leaf samples exhibited substantial changes in reflectance at most wavelengths as they dried that were strongly nonlinear with the stage of drying (Fig. 1). At visible wavelengths, leaf reflectance decreased sharply before leaf turgor loss, after which reflectance changes plateaued or even reversed sign. However, at near- and

shortwave-infrared wavelengths, leaf reflectance increased throughout the entire drydown and increased more rapidly with desiccation after TLP. Chow tests centered on TLP yielded mixed results: A large proportion of tests indicated that TLP represents a statistically significant breakpoint in how leaf reflectance responds to leaf drying, but the wavelengths with strong breakpoints varied substantially across and even within species (Fig. S6). In general, spectral responses to drying were strongly species-specific in ways not well explained by traits measured here (Fig. S7). Nearly all wavelengths yielded extremely high conditional R^2 values (mean cond. $R^2 = 0.983$) and only weak to moderate marginal R^2 values (mean marg. $R^2 = 0.292$) (Fig. S8). Most traits used in M2 were not statistically significant effects, excluding LMA and SWC, which were both statistically significant at many wavelengths and correlated with one another (Figs 2, S8). Collectively, our results suggest that the relationship between leaf water and leaf spectral reflectance varies nonlinearly with cell water status, strongly across species but also somewhat across samples, and is only weakly moderated by leaf water relations traits measured here.

In general, $NDVI_{leaf}$ increased as leaves dried, a pattern that is counterintuitive when considering most applications of NDVI at the ecosystem scale, such as tracking ecosystem health or estimating evapotranspiration (Goulden & Bales, 2014) (Fig. 3a). In line with the species-specific reflectance changes observed at the relevant wavelengths, this inverse relationship between LWA_{leaf} and $NDVI_{leaf}$ was strongly species-specific, with all models yielding very low marginal R^2 values and high conditional R^2 values and the majority of the variance being explained by the species random effect (Table 2). However, when considering the dataset in aggregate, we observed a positive relationship between LWA_{leaf} and $NDVI_{leaf}$. Counterintuitively, this means that when

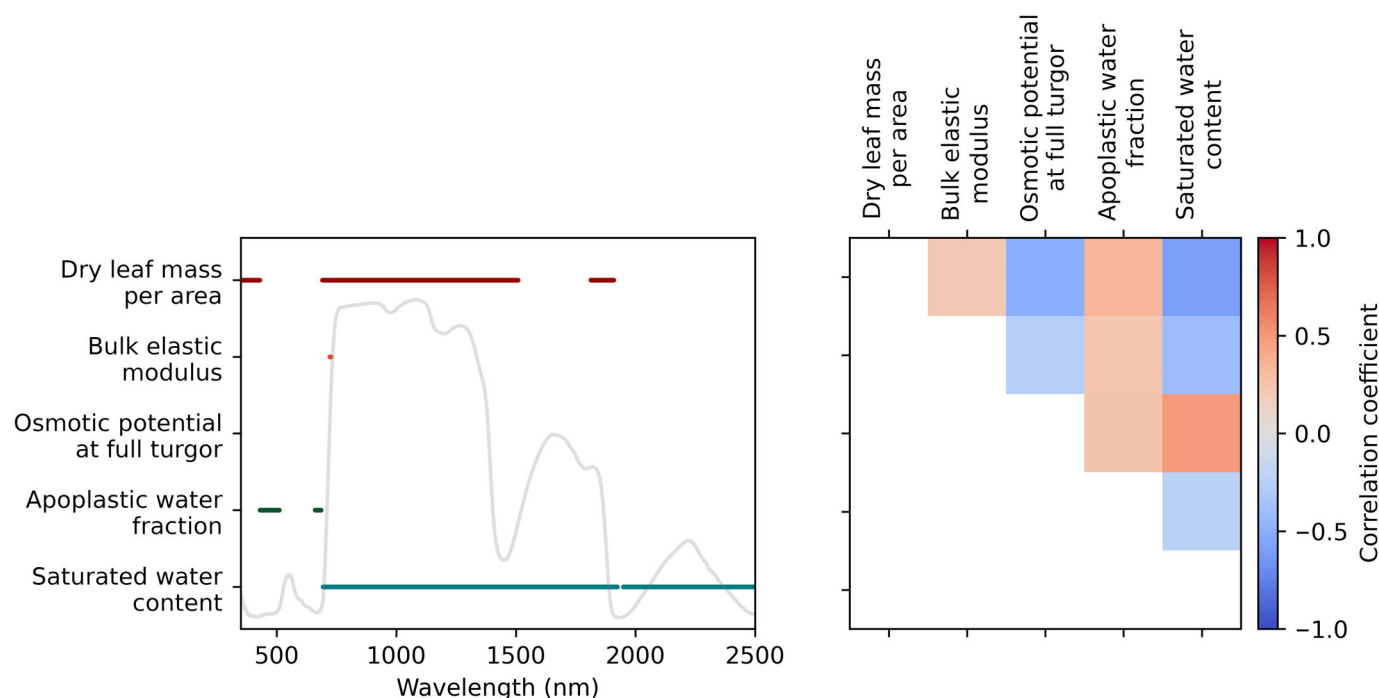


Fig. 2 Spectral responses to leaf drying are weakly mediated by dry leaf mass per area and saturated water content. Left: colored dots show wavelengths where each trait made a statistically significant ($P < 0.05$) contribution to the reflectance in M2 at that wavelength. In light gray, an arbitrary example spectrum is shown for reference. Right: correlation matrix of input variables. For a list of species considered in this analysis, see Table 1.

sampling many leaves from several species, a higher $\text{NDVI}_{\text{leaf}}$ is generally associated with higher LWA_{leaf} , but for multiple measurements from an individual leaf or an individual species, a higher $\text{NDVI}_{\text{leaf}}$ indicates lower LWA_{leaf} .

Finally, EWT_{leaf} and $\text{NDMI}_{\text{leaf}}$ both had strong and linear relationships with LWA_{leaf} , both when considering measurements from an individual leaf and across the whole dataset (Fig. 3b,c). Changes in EWT_{leaf} were strongly associated with changes in LWA_{leaf} , with M1 yielding the highest marginal R^2 of any model that did not leverage measured leaf traits (Table 2). Following the pattern established by previous models, these relationships were somewhat species-specific and leaf traits contributed only modestly to the variance explained. While M2 models (i.e. models including water relations traits) yielded notably higher marginal R^2 values than M0 models, species and sample random effects still explained roughly 20% of the variance (difference between marginal and conditional R^2 was 22 percentage points for EWT_{leaf} , 18 percentage points for $\text{NDMI}_{\text{leaf}}$). Interestingly, while the relationship between EWT_{leaf} and LWA_{leaf} was strong and linear, it was not one-to-one; EWT_{leaf} produced measurements that were, on average, nearly an order of magnitude larger than LWA_{leaf} . This is a substantially larger bias than observed in other studies (e.g. Allen *et al.*, 1971). Changes in EWT_{leaf} as the leaves dried were also fairly subtle, and declines in LWA_{leaf} from near saturation to well past TLP were typically associated with changes in EWT_{leaf} of a similar magnitude to the interspecies variability already present in the dataset.

Altogether, our datasets show that spectral responses to drying are highly species-specific, not only in the raw reflectance data but in spectral indices and in EWT_{leaf} . Across all analyses, these species-specific spectral responses were only somewhat mediated by the water relations traits measured here or by dry LMA, indicating that other factors not measured here are influencing how changes in LWA_{leaf} affect leaf spectral reflectance.

Canopy structural changes modulate the relationship between LWA and EWT

Our leaf-level spectroscopy results yielded species-specific leaf spectral responses to drying that have unidentified origins and are thus likely not represented in PROSAIL, which poses challenges to accurately simulating full-spectrum canopy-level responses to drying. For example, PROSPECT (the leaf spectroscopy component of PROSAIL) includes an EWT_{leaf} parameter meant to simulate drying, and indeed when inverting PROSPECT for the full spectra collected in this dataset, we do see that the EWT_{leaf} parameter in PROSPECT declines as leaves dry. However, several other PROSPECT parameters (e.g. LMA) change consistently as leaves dry in ways that are unrealistic (Fig. S9), indicating some level of mismatch between PROSPECT and our spectral measurements that could propagate forward to create error in our canopy-level, SAIL simulations. To overcome this, we chose to invert PROSPECT using only the wavelengths used when calculating EWT (1120–1280 nm), a common approach

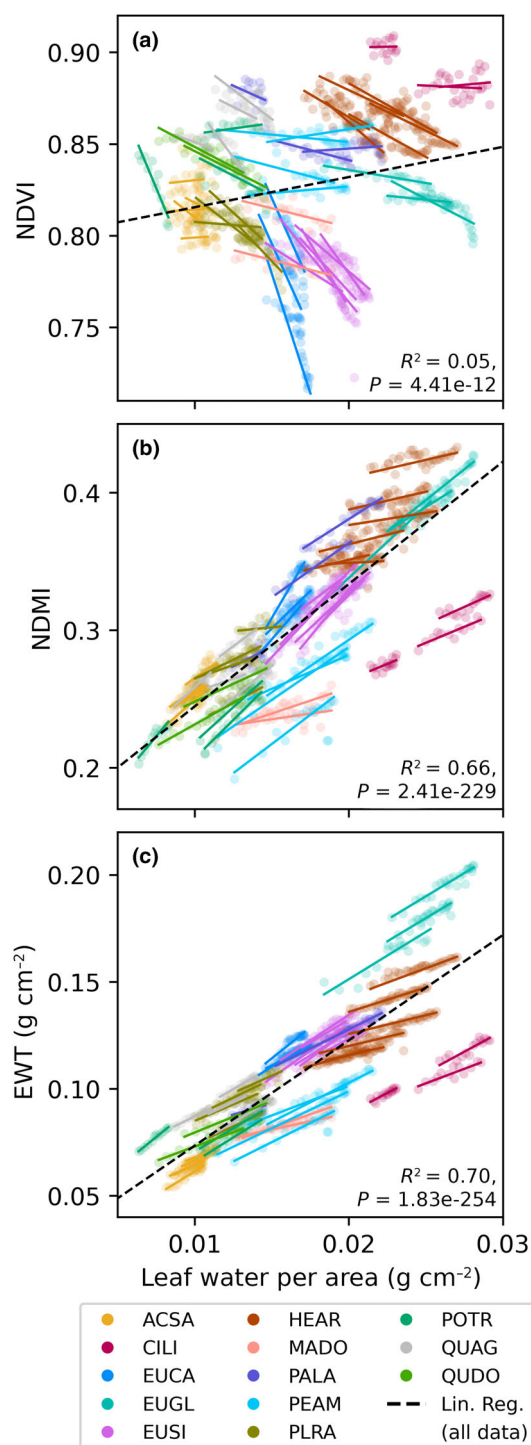


Fig. 3 The relationship between leaf water per area (LWA_{leaf}) and both equivalent water thickness (EWT_{leaf}) and $NDMI_{leaf}$ was positive, but varied across species; by contrast, the relationship between LWA_{leaf} and $NDVI_{leaf}$ was negative for most species, but positive across samples. (a) Each colored line here shows the linear regression between LWA_{leaf} and $NDVI_{leaf}$ for a single leaf sample, colored by that leaf's species. Definitions for each four-letter species code used in the legend can be found in Table 1. The black dotted lines and reported R^2 and P -value show linear regressions across the entire dataset but do not account for species or sampling. (b, c) Same as (a), but showing regressions between LWA_{leaf} and $NDMI_{leaf}$ and LWA_{leaf} and EWT_{leaf} , respectively. For a list of species considered in this analysis, see Table 1.

with PROSPECT to maximize accuracy at specific wavelengths (Féret *et al.*, 2019). When using these specific wavelengths, we observed very strong agreement between the EWT_{leaf} values calculated from the raw reflectance data and as represented in PROSPECT (mean difference = 0.002 g cm^{-2}). However, this focus on wavelengths in the 1120–1280 nm range limits our ability to accurately simulate canopy reflectance at other wavelengths (and therefore, to consider canopy-level changes in reflectance, $NDVI_{canopy}$, $NDMI_{canopy}$, or other spectra-derived values). For that reason, in this section, we focus exclusively on EWT_{canopy} .

Our modeling framework was able to successfully replicate several patterns that follow from existing observations and intuition, providing confidence that the simulated EWT_{canopy} measurements can serve as an adequate proxy for real data. Simulated EWT_{canopy} data showed moderate agreement with the actual canopy foliar water per unit area, LWA_{canopy} (Fig. S10): As expected, EWT_{canopy} increased with increasing LAI and LWA_{leaf} , though this relationship varied somewhat across species and also somewhat with the hotspot parameter q . Furthermore, mirroring observations at the leaf level, EWT_{canopy} biased high compared to LWA_{canopy} , an effect that has been observed in field-based studies (Bohn *et al.*, 2020). The magnitude of this bias declined with increasing LAI, pointing to a nonlinear relationship between EWT_{canopy} and LAI, which has also been observed in the field (Roberts *et al.*, 2004; Bohn *et al.*, 2020). Finally, EWT_{canopy} values simulated in our modeling framework yielded a similar range of values to EWT_{canopy} measurements made during a real imaging campaign (Fig. S11). Altogether, this agreement between the model outputs and real-world benchmarks indicates that our simulations should provide a realistic representation of canopy drydowns, or at minimum, accurately represent the key processes.

Leaf hydration and canopy structure both contributed substantially to simulated measurements of EWT_{canopy} (Fig. 4). In our leaf drydown experiment, the simulated change in EWT_{canopy} from water-saturated leaves to TLP was greater for higher LAI canopies – this follows intuition, since for higher LAI canopies, the same decline in LWA_{leaf} would result in a larger decline in LWA_{canopy} . However, changes in mean leaf angle simulated in our structural drydown experiment had diverging effects depending on the LAI: At low LAI, as average leaf angle increased by 15° (i.e. leaves became more erectophile), EWT_{canopy} declined; at higher LAI values, the same change in leaf angle resulted in a higher EWT_{canopy} , with larger changes observed at higher LAI.

This interaction between LAI and leaf angle means that the relative contributions of leaf structural changes (i.e. wilting) and leaf drying to EWT_{canopy} varied as a function of LAI. At low LAI values, changes in EWT_{canopy} are primarily explained by leaf drying, whereas at high LAI values, wilting substantially modulates the relationship between EWT_{canopy} and LWA_{canopy} . As leaf angle changes associated with water stress can be highly species-specific, the way that these competing factors (wilting and leaf drying) would contribute to an EWT_{canopy} measurement will vary substantially depending on the starting leaf angle as well as the leaf angle changes associated with wilting.

Table 2 Conditional R^2 , marginal R^2 , and SD of random effects (species, ID, and residual) for all models run using leaf equivalent water thickness (EWT_{leaf}), $NDVI_{leaf}$, and $NDMI_{leaf}$ as response variables.

Response variable	Model number	Cond. R^2	Marg. R^2	Sp. SD	ID SD	Resid. SD
EWT_{leaf}	M0	0.989	0.475	—	0.0169	0.00248
EWT_{leaf}	M1	0.789	0.686	—	0.0018	0.00248
EWT_{leaf}	M2	0.993	0.770	0.0130	0.0043	0.00248
$NDVI_{leaf}$	M0	0.974	0.150	—	0.0458	0.00820
$NDVI_{leaf}$	M1	0.812	0.087	—	0.0161	0.00820
$NDVI_{leaf}$	M2	0.968	0.112	0.0402	0.0129	0.00820
$NDMI_{leaf}$	M0	0.975	0.422	—	0.0315	0.00662
$NDMI_{leaf}$	M1	0.665	0.434	—	0.0055	0.00662
$NDMI_{leaf}$	M2	0.986	0.806	0.0223	0.0093	0.00662

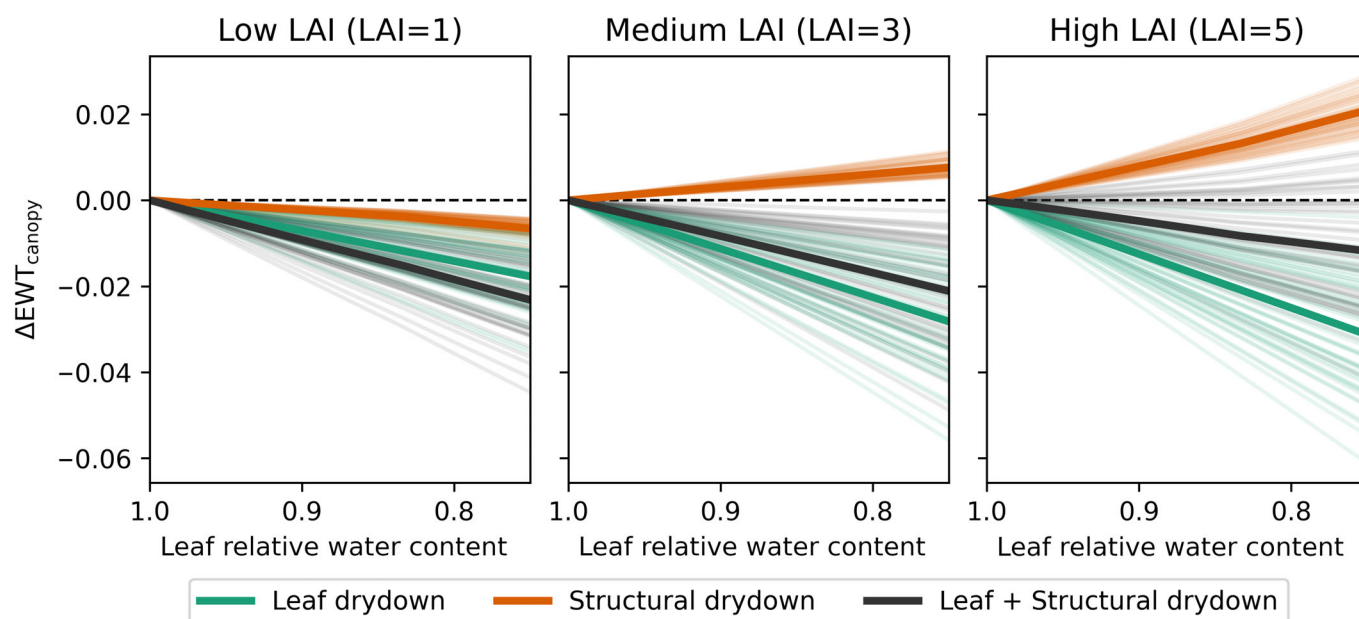


Fig. 4 As leaves dry, equivalent water thickness (EWT_{canopy}) (as simulated at the canopy level by PROSAIL) decreases; however, the magnitude of these changes in EWT_{canopy} is modified or offset by changes in leaf angle distribution (termed structural drydown) that are expected to occur simultaneously, and this effect varies in magnitude and direction depending on leaf area index (LAI). Translucent lines each represent one sample from the full dataset, whereas the opaque bold lines indicate the average across all samples. For a list of species considered in this analysis, see Table 1.

We also observed substantial variability in changes in EWT_{canopy} across species and samples. This is likely due to several interacting factors, including the starting LWA_{leaf} of the sample (which generally was correlated with LMA), species-specificity in the relationship between EWT_{leaf} and LWA_{leaf} , and the water use strategy of the tree. Generally, species with less negative TLP values and/or low capacitance (e.g. MADO and ACSA) exhibited smaller changes in LWA_{leaf} , and therefore EWT_{leaf} and EWT_{canopy} , but multiple counterexamples exist (e.g. EUGL and EUSI) (Fig. S12). Future research that leverages more species may be able to further disentangle what factors mediate this relationship.

Comparison with EWT error margins

Overall fit between surface reflectance before and after atmospheric simulations and inversions was good, with the most

accurate surface reflectance values retrieved for the more recently developed, lower noise instrument (EMIT) compared to the older instrument that has more sensor noise (AVIRIS-NG) (Fig. S13). The mean absolute error ranged from 0.8 to 1.4% reflectance depending on the instrument simulated. Previous research using versions of this surface reflectance retrieval algorithm for ground truth yielded mean absolute errors for vegetation of $\approx 1\text{--}2\%$ for the instrument sampled here with the higher noise (Thompson *et al.*, 2018); this indicates that our surface reflectance retrievals are overall good and on par with current operational retrievals. For AVIRIS-NG, we estimate that EWT shows a positive bias of $+0.006\text{ g cm}^{-2}$ and a precision (SD of measurement errors) of 0.019 g cm^{-2} per observation. EMIT exhibits less bias and higher precision, with a negative bias of -0.002 g cm^{-2} and a precision of 0.014 g cm^{-2} per observation.

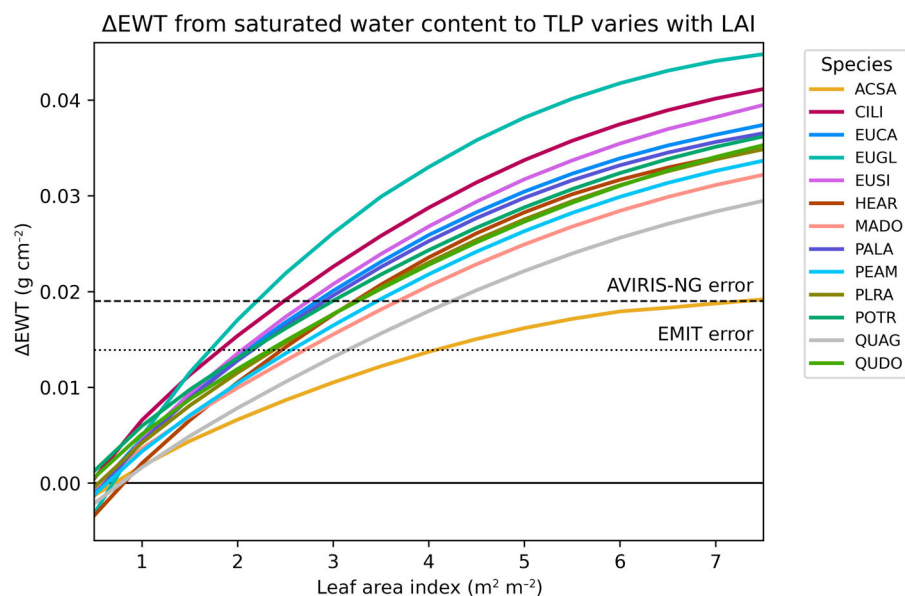


Fig. 5 Changes in equivalent water thickness (EWT_{canopy}) simulated by PROSAIL during a combined leaf and structural drydown (i.e. a realistic drydown) from full saturation to turgor loss point (TLP) increased with leaf area index (LAI) and varied across species, and were of similar magnitude to the error associated with EWT_{canopy} retrieval for each of the two instruments simulated here. Values on the y-axis indicate the change in EWT_{canopy} simulated based on a linear regression between relative water content and EWT_{canopy} , and colors indicate averages across all samples for each species. Definitions for each four-letter species code used in the legend can be found in Table 1. For a list of species considered in this analysis, see Table 1.

However, for both AVIRIS-NG and EMIT, these estimates of EWT_{canopy} error are of similar magnitude to the EWT_{canopy} changes observed in ‘Canopy structural changes modulate the relationship between LWA and EWT’ in the Results section associated with changes in tree water status (Fig. 5). Using the instrument error metrics described above, for a single pixel, declines in leaf water from saturation to TLP were only detectable at a 95% confidence level for moderate-to-high LAI canopies (avg. min. LAI across species = $4.91 \pm 1.23 \text{ m}^2 \text{ m}^{-2}$ for EMIT, $> 8 \text{ m}^2 \text{ m}^{-2}$ for AVIRIS-NG). This average minimum required LAI for a single-pixel retrieval is more than twice the global average (Fang *et al.*, 2019) and substantially exceeds the typical LAI observed among several of our study species (Casella & Ceulemans, 2002; White *et al.*, 2010; Allen *et al.*, 2026). In practice, remote sensing analyses that leverage only a single pixel are rare, as remote sensing instruments typically collect thousands of measurements per scene. As a result, these large retrieval errors may not be prohibitive to the use of EWT_{canopy} for detecting canopy drydowns, and this means that this ‘single-pixel’ LAI threshold is an extremely conservative figure when considering downstream applications. However, even when EWT_{canopy} measurements from multiple pixels are averaged together (assuming a homogeneous canopy and uncorrelated error between pixels), the average needed LAI remains moderately high. For a 3×3 pixel kernel containing nine identical, independent measurements, the average minimum needed LAI to detect a full drydown at 95% confidence is $1.68 \pm 0.37 \text{ m}^2 \text{ m}^{-2}$ for EMIT and $2.18 \pm 0.52 \text{ m}^2 \text{ m}^{-2}$ for AVIRIS-NG. More subtle drydowns (e.g. from full saturation to half-TLP, medium-LAI canopies that do not exhibit any wilting, etc.) induce smaller changes in EWT_{canopy} and therefore require more measurements to be observed with confidence, either in time or across space.

Discussion

In this paper, we collected spectral measurements of leaf drying and quantified changes in leaf reflectance over the course of leaf

drydowns for 13 broadleaf tree species. We found that changes in leaf reflectance that occurred over a drydown varied across species and pre- vs post-turgor loss. Both across and within species, EWT_{leaf} tracked changes in LWA_{leaf} substantially better than $NDVI_{\text{leaf}}$ and modestly better than $NDMI_{\text{leaf}}$ or reflectance at any given band, though absolute values of EWT_{leaf} and LWA_{leaf} were not directly comparable. Next, in order to compare the relative effects of leaf spectral and canopy structural changes (as quantified through changes in leaf angle) in contributing to an EWT_{canopy} measurement, we used a radiative transfer modeling framework to simulate measurements of EWT_{canopy} observed by an airborne or spaceborne spectrometer. We found that canopy structural changes associated with water stress (i.e. increasing average leaf angle by 15° , simulating wilting) can cause diverging responses in EWT_{canopy} at low vs high LAI values. This leaf angle effect could therefore either amplify (at low LAI) or counteract (at high LAI) declines in EWT_{canopy} associated with declining leaf water content. Finally, by comparing our simulated changes in EWT_{canopy} to error values calculated using realistic simulations of reflectance retrievals, we found that EWT_{canopy} error values for modern-day instruments are non-negligible compared to EWT_{canopy} changes associated with tree water stress, and that EWT_{canopy} is expected to only be sensitive to the effects of water stress when leveraging many measurements in aggregate that are assumed to be otherwise identical (e.g. same species, same canopy structural response, etc.).

Species-specific leaf-level responses to drying

Of the spectral response variables considered in this paper, EWT_{leaf} was the most strongly related to LWA_{leaf} , both within and among individuals. Although we cannot definitively partition the large among-individual variation to variation among species vs among individuals within a species with our sample sizes, it likely represents variation among species (Fig. S2). EWT_{leaf} (as calculated using the Beer–Lambert law) may seem somewhat

reductive as a model: It does not account for the scattering of light within leaves in any way, let alone in ways that accurately capture the diverse and complex optical dynamics of real leaves (Gonzalez *et al.*, 2025). Despite this, we still see good agreement between EWT_{leaf} and LWA_{leaf} , particularly within a sample or species, but also when considering taxonomically diverse measurements. This indicates that – in many potential use cases – EWT_{leaf} is an extremely helpful tool for assessing leaf hydration from spectroscopy data. Calculating EWT_{leaf} requires no training data or trait measurements, is not especially computationally burdensome, and – since the coefficients of absorption of liquid water are well-known and publicly available (Kedenburg *et al.*, 2012) – the only limitation to measurement accuracy is instrumentation. Perhaps most importantly, EWT_{leaf} also has a directly comparable canopy level measurement (EWT_{canopy}), providing studies that use it with a natural opportunity to scale their leaf level observations to hypotheses about dynamics that might be observed in real imagery. These are major advantages.

However, while most spectral response variables (except $NDVI_{leaf}$) were positively related to LWA_{leaf} to some degree, substantial species-specificity was observed in the relationship that was not well explained by differences in the traits measured here. This interspecies variability was greater than the within species variability in all cases. One might hypothesize that a more complex model is able to account for these differences. However, PROSPECT only partially captured the diverse spectral responses to drying we observed here, and often required unrealistic parameter adjustments, such as LMA changing dynamically throughout the drydown (Fig. S9). Furthermore, PROSPECT's EWT_{leaf} parameter yielded species differences that were only marginally smaller than those seen with the simpler Beer–Lambert EWT_{leaf} definition used throughout this paper (Fig. S14). This species discrepancy, only somewhat represented through traits measured here and mechanisms implemented in PROSPECT, points to a gap in our understanding of how leaf reflectance changes as leaves dry, with practical implications for studies that use spectroscopy to phenotype plants or measure traits; leaf hydration should be controlled or explicitly represented in predictive models or it will likely serve as a confounding factor. An interesting pattern that emerges is that our three agricultural tree species (MADO, PEAM, and CILI) all yield lower EWT_{leaf} values at a given LWA_{leaf} value, for both the low-LMA and high-LMA samples. One hypothesis is that these cultivated trees, which are not subject to the same selection pressures as naturally occurring trees, have leaf anatomical differences (e.g. lengths of the palisade cells, vein architecture) that cause near-infrared light to have a shorter path length through the leaf before being reflected back out for measurement by a sensor.

For direct measurement of leaf hydration, the confounding effect of species differences is largely mitigated for EWT_{leaf} by modeling it as a change from a baseline rather than as an absolute value. This means that, while absolute measurements of EWT_{leaf} can only be moderately linked to specific LWA_{leaf} values without a model calibrated for those specific leaf traits or that specific species, changes in EWT_{leaf} are directly informative of changes in LWA_{leaf} . As was also seen in the canopy simulations, this

indicates that timeseries of EWT_{leaf} over periods of changing leaf hydration are more useful than single measurements, not only because they provide more data to leverage but also because the differences themselves can be more reliably linked to changing LWA_{leaf} , even if the time series only contains two points. According to our data, detecting changes in EWT_{leaf} and/or EWT_{canopy} over time largely removes the need for species-specific models.

Given the strong history of $NDVI_{canopy}$ for use in assessing plant health (e.g. Chávez *et al.*, 2016; Xue & Su, 2017; Stamford *et al.*, 2023), the observed increases in $NDVI_{leaf}$ associated with leaf drying may seem counterintuitive. However, this finding does not contradict previous research. $NDVI_{canopy}$ is typically applied to long time series of data and/or large regions with diverse species compositions and canopy structures, where differences in LAI and species distribution/turnover likely dwarf hydration-related changes in leaf spectra. The body of research concerning $NDVI_{leaf}$ is small but overall supports our findings. Multiple studies have observed changes in near-infrared reflectance and little to no change in red reflectance over the course of a leaf drydown, which, if $NDVI_{leaf}$ had been calculated, would result in the inverse relationship observed here (Labovitz *et al.*, 1984; Carter, 1991; Cibula *et al.*, 1992; Yu *et al.*, 2000; Chávez *et al.*, 2016; Xue & Su, 2017; Stamford *et al.*, 2023). There have been some papers that observed a positive relationship between $NDVI_{leaf}$ and LWA_{leaf} (Seelig *et al.*, 2008), but these studies used several leaves from different individual trees or different species, likely detecting the positive among-leaf or among-species relationship we found in Fig. 3a. Our results suggest that, in most cases, any observed sensitivity of $NDVI_{canopy}$ to LWA_{leaf} is likely not due to a leaf-level relationship but rather to species composition changes, leaf senescence or abscission, or canopy structural changes.

EWT_{canopy} responds to both structural and leaf-level changes induced by drought

EWT_{canopy} is often described as a measurement of water contained in vegetation per unit area; however, here we observed that the link between EWT_{canopy} and LWA_{canopy} is more complicated than may be expected. For a given canopy structure (holding LAI and leaf angle constant), EWT_{canopy} and LWA_{canopy} yielded a relationship similar to that observed between EWT_{leaf} and LWA_{leaf} , with overall agreement and some species-specificity. However, across varying canopy structures, this relationship became substantially more complex: EWT_{canopy} overestimated LWA_{canopy} in all simulations, but the magnitude of this bias varied nonlinearly with LAI. Our results suggest that studies that sample from structurally homogeneous canopies would likely observe a positive relationship between LWA_{canopy} and EWT_{canopy} , whereas studies that span greater canopy structural heterogeneity may indeed observe no relationship, even in cases where that heterogeneity appears subtle. Likewise, the temporal relationship between LWA_{canopy} and EWT_{canopy} is complicated by concurrent changes in canopy structure, which may explain the nonexistent relationships between EWT_{canopy} and leaf water content and even directly calculated LWA_{canopy} in airborne

spectroscopy studies (Martin *et al.*, 2018; Boving *et al.*, 2025). Future imaging spectroscopy studies that aim to relate LWA_{canopy} and EWT_{canopy} – especially those that cover large and diverse study areas – will need to explicitly account for this structural effect, either through canopy structural measurements (e.g. lidar) or careful selection of data (e.g. timeseries of well-known study sites).

We found a consistent overestimation of leaf water by EWT_{leaf} and EWT_{canopy} . This creates opportunities for more precise monitoring of changes in leaf water that are effectively amplified in remotely sensed data. Long optical path lengths of light through leaves and canopies amplify the changes in both EWT_{leaf} and EWT_{canopy} observed during a leaf drydown when compared to LWA_{leaf} or LWA_{canopy} , substantially improving the relative precision of EWT measurements. It is also likely that species differences in how water limitation influences plant structure, such as leaf angle changes, would exacerbate differences in EWT_{canopy} changes. While it is widely speculated and understood that leaf angle adjustments and changes in LAI (e.g. leaf shedding) in response to drought are likely highly diverse across taxonomic and trait axes, there exists very little published data on this diversity, and the 'structural drydown' simulated here represents only one example of how EWT_{canopy} may change in response to canopy structural adjustments. Further field measurements of how leaf angle responds to plant water stress would substantially aid in our ability to leverage EWT_{canopy} measurements.

Implications for mapping tree water stress at the landscape scale

The high EWT_{canopy} retrieval error estimates quantified by our simulations relative to the magnitude of the canopy drying signal present considerable challenges for both validation of EWT_{canopy} data products and application of EWT_{canopy} for monitoring fluctuations in leaf water. The high EWT_{canopy} error estimate made here is paralleled by similar findings for the German Environmental Mapping and Analysis Program (EnMAP) instrument (Bohn *et al.*, 2020). For both instruments assessed in this study, we found that the EWT_{canopy} signal of leaf drying from full saturation to turgor loss only surpasses retrieval error for canopies with quite high LAI values. Furthermore, for AVIRIS-NG, the error was higher when EWT_{canopy} was low (Fig. S13). Therefore, in low LAI, dry ecosystems, we could expect the noise to obscure the leaf drying signal for EWT_{canopy} , making drydowns extremely difficult to detect, barring exceptionally large data volumes. At broad scales, hyperspectral datasets generally contain thousands of measurements that can offset high EWT_{canopy} retrieval errors; unfortunately, this makes validating EWT_{canopy} data products with point-based field measurements a massive undertaking. Thus, using EWT_{canopy} for monitoring fluctuations in leaf water is something of a black box – if we assume this method works, then we have the necessary data volumes to apply it, but it may be a long time before we can know for sure.

Here, we attempted to simulate a reasonable range of vegetation states. However, ecosystems are complex systems and there are numerous properties and structural complexities not represented in SAIL or other current vegetation radiative transfer

models. At present, the data needed to externally validate our results do not exist: Subseasonal measurements of EWT_{canopy} are extremely rare, and we are not familiar with any high-frequency VSWIR imaging campaign that also collected coincident data on canopy structure for a sufficiently wide array of tree species. However, the upcoming NASA Explorer for Artemis Geology Lunar and Earth (EAGLE-VSWIR) mission will provide an opportunity to validate the results identified in our model framework. EAGLE-VSWIR will collect global VSWIR data at 30 m spatial resolution every 16 d, providing high-frequency measurements of EWT_{canopy} across a wide variety of ecosystems and climates, spanning a host of different species, LAI, and canopy structural gradients. Time series measurements of EWT_{canopy} made by EAGLE-VSWIR will capture drought events in action in diverse ecosystems across the globe. These data will provide opportunities for further research into how the relationship between EWT_{canopy} and LWA_{canopy} varies across species, systems with different LAIs, and across space and time with differences in drought intensity and duration. From a model validation perspective, our modeling framework predicts that leaf-level and canopy-level EWT biases should be correlated, with certain species (e.g. EUGL) producing disproportionately large EWT measurements at both the leaf and canopy level. By comparing the cross-species relationship between EWT_{canopy} (measured by EAGLE-VSWIR) and LWA_{canopy} to the cross-species relationship between EWT_{leaf} and LWA_{leaf} , we should be able to verify this finding. Furthermore, the recently announced Earth Dynamics Geodetic Explorer (EDGE), a spaceborne lidar mission slated for launch in 2030 or later, may also resolve one of the key uncertainties of this study by characterizing tree structural responses to drought, which are currently poorly constrained. EDGE will provide opportunities to both characterize tree structural responses to drought across species and compare them with changes in EWT_{canopy} across ecosystem types. These key validations would open the door to more informed application of EWT_{canopy} data at a global scale and enable more robust mapping of tree water stress at relatively high spatial resolutions.

Conclusion

Our results suggest that ideal use cases for hyperspectral imaging for water stress mapping would be large, contiguous vegetated areas with high LAI and a single dominant overstory species. In these ecosystems, changes in EWT_{canopy} associated with a leaf drying event would be relatively large and could exceed error margins for the EWT_{canopy} measurement. Furthermore, since canopy structural effects exhibit nonlinear and large effects on EWT_{canopy} , and many species exhibit different structural responses to drought, controlling for species may also allow for controlling of these effects. Finally, if most or all of the leafy biomass across individual trees is expected to be exhibiting similar leaf water fluctuations, this would increase the observed change in EWT_{canopy} as well as allow for aggregation of EWT_{canopy} measurements from multiple pixels, reducing the SE of the measurement and increasing the statistical confidence with which a change in EWT_{canopy} associated with a leaf drying can be

detected. The greater consistency between EWT_{canopy} and leaf water observed in agricultural literature is aligned with this suggested ideal use case. Absent these ideal conditions, it may alternatively be possible to overcome the high EWT_{canopy} retrieval errors by a dense time series of EWT_{canopy} measurements for a single pixel or single region.

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Competing interests

None declared.

Author contributions

JA, LDLA, DR and ATT planned and designed the research. JA performed experiments and analyzed data. JA and ATT wrote the manuscript. JA, LDLA, DR and ATT revised the manuscript.

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Data availability

The data that support the findings of this study are openly available on EcoSIS (doi: [10.21232/egGynzX](https://doi.org/10.21232/egGynzX)), and all code used for analysis and figure generation is available on GitHub at https://github.com/jean-allen/spectral_pv_curves.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Comparison between mean turgor loss point and leaf mass per area across species.

Fig. S2 Comparison between intraspecific and interspecific spectral responses to drying.

Fig. S3 Bar chart showing initial water potential values for all samples.

Fig. S4 Annotated lab setup.

Fig. S5 Shows how EWT_{canopy} error estimates vary with atmospheric/viewing conditions.

Fig. S6 Chow test results for turgor loss point as a breakpoint across species and wavelengths.

Fig. S7 Interspecies vs intraspecies variability in spectral responses to drying.

Fig. S8 Marginal and conditional R^2 values for mixed-effects models as a function of wavelength.

Fig. S9 Examples of unrealistic drying dynamics in PROSPECT parameters.

Fig. S10 Comparison between ‘true’ EWT_{canopy} values and simulated EWT_{canopy} measurements.

Fig. S11 Comparison between ‘true’ EWT_{canopy} values and simulated EWT_{canopy} measurements.

Fig. S12 Comparison between leaf water relations traits and EWT_{canopy} .

Fig. S13 Input vs output plot for EWT_{canopy} error calculation.

Fig. S14 Comparison between LWA_{leaf} and the EWT parameter used in PROSPECT.

Notes S1 Detailed description of EWT_{canopy} error calculation.

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